

METHOD – I

LDA-Guided PCA Fusion for Multi-Modal LiDAR Imaging

1. Problem Setup

Each pixel in the LiDAR dataset is represented as a **4-dimensional feature vector**:

$$x = [x_1, x_2, x_3, x_4]^T \in \mathbb{R}^4$$

corresponding to:

- reflectivity
- signal intensity
- near-infrared
- range

Using bounding box annotations, we assign binary labels:

$$y \in \{0,1\}$$

where:

- $y = 1$: foreground (snow pole)
- $y = 0$: background

2. Linear Discriminant Analysis (LDA)

Objective

We first seek a **projection direction** $w \in \mathbb{R}^4$ that maximizes class separability. LDA achieves this by maximizing the **Fisher criterion**:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

Scatter Matrices

Let:

- μ_1, μ_0 : class means
- S_W : within-class scatter

- S_B : between-class scatter

$$S_B = (\mu_1 - \mu_0)(\mu_1 - \mu_0)^T$$

$$S_W = \sum_{i \in 1} (x_i - \mu_1)(x_i - \mu_1)^T + \sum_{i \in 0} (x_i - \mu_0)(x_i - \mu_0)^T$$

Solution

The optimal projection direction is:

$$w \propto S_W^{-1}(\mu_1 - \mu_0)$$

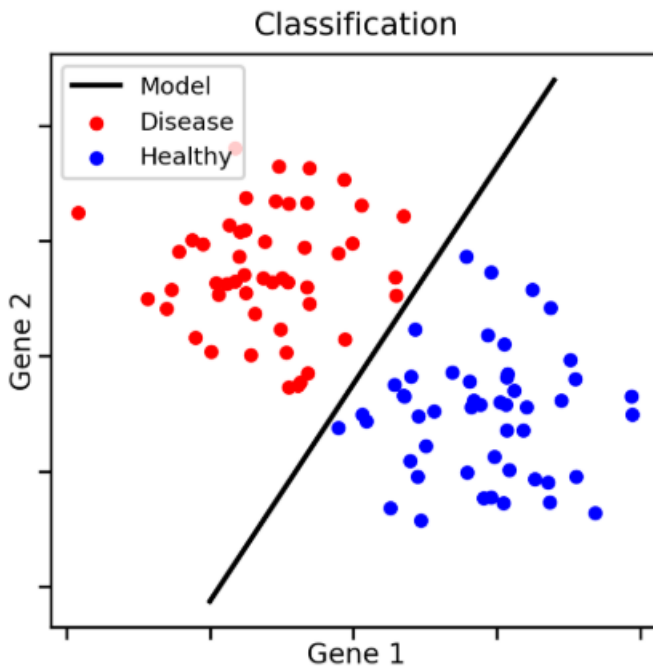
We normalize:

$$w \leftarrow \frac{w}{\|w\|}$$

Projection

Each pixel is projected onto this direction:

$$z_0 = w^T x$$



This scalar represents the **most discriminative feature** separating foreground and background.

3. Orthogonal Residual Decomposition

To avoid redundancy, we remove the component of x aligned with w :

$$x_{\text{res}} = x - (w^T x)w$$

Interpretation

This is a **Gram–Schmidt orthogonalization step**, decomposing the signal into:

$$x = \underbrace{(w^T x)w}_{\text{discriminative component}} + \underbrace{x_{\text{res}}}_{\text{orthogonal residual}}$$

Thus, x_{res} contains variation **independent of class discrimination**.

4. Principal Component Analysis (PCA) on Residual

We apply PCA to the residual data to capture dominant structure in the orthogonal subspace.

Covariance Matrix

Let centered residuals be:

$$\tilde{x}_i = x_{\text{res},i} - \mu_{\text{res}}$$

The covariance matrix is:

$$\Sigma = \frac{1}{N} \sum_{i=1}^N \tilde{x}_i \tilde{x}_i^T$$

Optimization Problem

PCA seeks directions maximizing projected variance:

$$\begin{aligned} u_1 &= \arg \max_{\|u\|=1} u^T \Sigma u \\ u_2 &= \arg \max_{\|u\|=1, u \perp u_1} u^T \Sigma u \end{aligned}$$

Solution via Eigen Decomposition

The optimal directions satisfy:

$$\Sigma u = \lambda u$$

- u_1 : eigenvector with largest eigenvalue λ_1

- u_2 : eigenvector with second largest eigenvalue λ_2

Projection

Residual features are projected as:

$$z_1 = u_1^T x_{\text{res}}, z_2 = u_2^T x_{\text{res}}$$

5. Final Feature Representation

Each pixel is mapped from $\mathbb{R}^4 \rightarrow \mathbb{R}^3$:

$$z = [z_0, z_1, z_2]^T$$

where:

- z_0 : discriminative (LDA) component
- z_1, z_2 : principal residual components

Compact Formulation

The transformation can be written as:

$$z = \begin{bmatrix} w^T \\ u_1^T(I - ww^T) \\ u_2^T(I - ww^T) \end{bmatrix} x$$

6. Normalization and Image Construction

Each channel z_k is normalized using robust percentile scaling:

$$\hat{z}_k = \frac{\text{clip}(z_k, p_1, p_{99}) - p_1}{p_{99} - p_1}$$

where:

- p_1, p_{99} : 1st and 99th percentiles computed over training data

The normalized channels are mapped to 8-bit range:

$$I_k = 255 \cdot \hat{z}_k$$

RGB Mapping

The final fused image is constructed as:

- **R channel:** z_0 (LDA)
- **G channel:** z_1 (PCA-1)
- **B channel:** z_2 (PCA-2)

7. Interpretation

This method constructs a **hybrid feature basis**:

- w : maximizes **class separability** (supervised)
- u_1, u_2 : maximize **data variance** (unsupervised)

Thus, the representation combines:

- **discriminative information**
- **structural information**

while ensuring orthogonality and minimal redundancy.

8. Key Properties

- **Dimensionality reduction:** $4 \rightarrow 3$
- **Orthogonal decomposition:** $w \perp u_1 \perp u_2$
- **Information efficiency:** preserves both label-driven and intrinsic variance

9. Summary

The proposed pipeline performs:

Supervised projection (LDA) followed by orthogonal unsupervised compression (PCA)

resulting in a compact, information-rich representation optimized for downstream object detection.

METHOD – II

1. Input Representation

Each sample consists of two aligned modalities:

$$x_c \in \mathbb{R}^{3 \times H \times W}, x_r \in \mathbb{R}^{1 \times H \times W}$$

2 Main stream (Comb4)

YOLO backbone acts as a function:

$$\phi: \mathbb{R}^{3 \times H \times W} \rightarrow \mathbb{R}^{C_c \times \frac{H}{16} \times \frac{W}{16}}$$

$$F_c = \phi(x_c)$$

where:

$$C_c = 128$$

3 Range stream (auxiliary encoder)

The range encoder is a composition of convolutional operators:

$$\psi = f_4 \circ f_3 \circ f_2 \circ f_1$$

Each layer:

$$f_k(x) = \sigma(\text{BN}(W_k * x))$$

So:

$$F_r = \psi(x_r)$$

$$F_r \in \mathbb{R}^{C_r \times \frac{H}{16} \times \frac{W}{16}}, C_r = 128$$

4.1 Concatenation

At each location:

$$f_c \in \mathbb{R}^{C_c}, f_r \in \mathbb{R}^{C_r}$$

$$f_{\text{concat}} = \begin{bmatrix} f_c \\ f_r \end{bmatrix} \in \mathbb{R}^{C_c + C_r}$$

Globally:

$$F_{\text{concat}} = \text{Concat}(F_c, F_r) \in \mathbb{R}^{256 \times \frac{H}{16} \times \frac{W}{16}}$$

4.2 Linear Projection (1×1 Convolution)

Fusion is performed via a learnable linear map:

$$W \in \mathbb{R}^{C_c \times (C_c + C_r)} = \mathbb{R}^{128 \times 256}$$

At each spatial location:

$$f_{\text{fused}} = W f_{\text{concat}}$$

Final fusion equation:

$$f_{\text{fused}} = \sigma(\text{BN}(W_c f_c + W_r f_r))$$

BN – Batch normalization

Fusion uses:

$$\sigma(x) = x \cdot \frac{1}{1 + e^{-x}}$$

i.e., SiLU (Swish) activation.

Output channels – 128

Then the input stream passes to the next layers

METHOD – III QM-PCA

Same as LDA-PCA's PCA, except that 4 channels are projected into 3.

METHOD – IV

Anisotropic Fusion

1. Input Representation

Each modality is a scalar field:

$$I_m(x, y), m \in \{\text{reflec}, \text{signal}, \text{nearir}, \text{range}\}$$

So we have **4 images**:

$$\{I_1, I_2, I_3, I_4\}$$

2. Normalization

Each modality is standardized:

$$\tilde{I}_m = \frac{I_m - \mu_m}{\sigma_m}$$

where:

- μ_m : mean intensity
- σ_m : standard deviation

3.1 Directional gradients

For each pixel (i, j) :

$$\begin{aligned}\delta_N(i, j) &= \tilde{I}_m(i - 1, j) - \tilde{I}_m(i, j) \\ \delta_S(i, j) &= \tilde{I}_m(i + 1, j) - \tilde{I}_m(i, j) \\ \delta_E(i, j) &= \tilde{I}_m(i, j + 1) - \tilde{I}_m(i, j) \\ \delta_W(i, j) &= \tilde{I}_m(i, j - 1) - \tilde{I}_m(i, j)\end{aligned}$$

$$\delta_N, \delta_S, \delta_E, \delta_W \in \mathbb{R}^{H \times W}$$

3.2 Conductance coefficients

$$c_N = \exp \left(- \left(\frac{\delta_N}{\kappa} \right)^2 \right)$$

(similarly for c_S, c_E, c_W)

$$c_N, c_S, c_E, c_W \in \mathbb{R}^{H \times W}$$

3.3 Diffusion update

At iteration t :

$$I_m^{(t+1)} = I_m^{(t)} + \gamma(c_N \odot \delta_N + c_S \odot \delta_S + c_E \odot \delta_E + c_W \odot \delta_W)$$

where:

- $\odot$: element-wise multiplication
- $I_m^{(t)} \in \mathbb{R}^{H \times W}$
- Update term $\in \mathbb{R}^{H \times W}$

3.4 Iteration

$$t = 1, 2, \dots, T$$

Final output:

$$\hat{I}_m = I_m^{(T)} \in \mathbb{R}^{H \times W}$$

4. Multi-Modal Fusion

You fuse all modalities via averaging:

$$I_{\text{fused}} = \frac{1}{4} \sum_{m=1}^4 \hat{I}_m$$

$$I_{\text{fused}} \in \mathbb{R}^{H \times W}$$

5. Min-Max Normalization

Let:

$$I_{\min} = \min_{i,j} I_{\text{fused}}(i,j)$$

$$I_{\max} = \max_{i,j} I_{\text{fused}}(i,j)$$

Normalize:

$$I' = \frac{I_{\text{fused}} - I_{\min}}{I_{\max} - I_{\min} + \epsilon}$$

$$I' \in \mathbb{R}^{H \times W}$$

Output – 1 channel Image

METHOD – V

Range Enhanced PCA Fusion

1. Transform range:

1 Gradients

$$G_x = \frac{\partial I}{\partial x}, G_y = \frac{\partial I}{\partial y}$$

2 Gradient magnitude(Sobel Filter)

$$I'_{\text{range}} = \sqrt{G_x^2 + G_y^2}$$

Input -

$$x = [x_1, x_2, x_3, x_4]^T \in \mathbb{R}^4$$

where:

- x_1 : reflectivity
- x_2 : signal
- x_3 : nearIR
- x_4 : enhanced range

2. Apply PCA

METHOD – VI

Early PCA and Residual

1 Dataset

$$X_{\text{train}} \in \mathbb{R}^{N \times 4}$$

2 Standardization

$$X_{\text{std}} \in \mathbb{R}^{N \times 4}$$

3 PCA decomposition

$$\Sigma u_k = \lambda_k u_k$$

This is the **core object PCA** uses

$$\Sigma = \frac{1}{N} X_{\text{std}}^T X_{\text{std}}$$

Dimensions

$$\Sigma \in \mathbb{R}^{4 \times 4}$$

It looks like:

$$\Sigma = \begin{bmatrix} \text{cov}(x_1, x_1) & \cdots & \text{cov}(x_1, x_4) \\ \vdots & \ddots & \vdots \\ \text{cov}(x_4, x_1) & \cdots & \text{cov}(x_4, x_4) \end{bmatrix}$$

◆ **Step 3: Eigen Decomposition (THIS gives u and λ)**

You solve:

$$\Sigma u_k = \lambda_k u_k$$
$$\lambda^4 + a_1 \lambda^3 + a_2 \lambda^2 + a_3 \lambda + a_4 = 0$$

◆ **What this produces:**

(A) u_k

- A vector:

$$u_k \in \mathbb{R}^4$$

Example:

$$u_1 = [0.6, 0.5, 0.4, 0.3]$$

This is one PCA direction

(B) λ_k

- A number:

$$\lambda_k \in \mathbb{R}$$

tells how important that direction is

So:

- u_k = direction (how modalities combine)
- λ_k = strength of that direction

So:

$$U = \begin{bmatrix} u_1^T \\ u_2^T \\ u_3^T \\ u_4^T \end{bmatrix} \in \mathbb{R}^{4 \times 4}$$
$$\lambda = [\lambda_1, \lambda_2, \lambda_3, \lambda_4]$$

Contribution score, Compute:

$$c_j = \sum_{k=1}^4 \lambda_k \cdot |u_{k,j}|$$

Quantity Shape

$$X \qquad N \times 4$$

$$\Sigma \qquad 4 \times 4$$

$$u_k \qquad 4$$

$$\lambda_k \qquad \text{scalar}$$

$$c \qquad 4$$

$$w{=} [0.2485,0.2555,0.2320,0.2640]$$

4. Low-Frequency Fusion (Weighted Sum)

For each pixel:

$$I_{\text{LF}}(i,j) = \sum_{m=1}^4 w_m \tilde{I}_m(i,j)$$

$$I_{\text{LF}} \in \mathbb{R}^{H \times W}$$

5. High-Frequency Edge Residual

Normalize inputs

$$\begin{aligned} R &= \text{range}, F = \text{reflec} \\ R', F' &\in [0,1]^{H \times W} \end{aligned}$$

Gradients

$$\nabla R = (R_x, R_y), \nabla F = (F_x, F_y)$$

Magnitude

$$G_R = \sqrt{R_x^2 + R_y^2}, G_F = \sqrt{F_x^2 + F_y^2}$$

Edge fusion

$$E = 0.5 \cdot G_R + 0.5 \cdot G_F$$

5 Normalize

$$E \in [0,1]^{H \times W}$$

6. Final Fusion

$$I_{\text{fused}} = I_{\text{LF}} + \gamma E$$

where:

- $\gamma = 0.05$

$$I_{\text{fused}} \in \mathbb{R}^{H \times W}$$

7. Final Normalization

$$I' = \frac{I_{\text{fused}} - \min}{\max - \min}$$
$$I_{\text{final}} = 255 \cdot I'$$

Final output

$$I_{\text{final}} \in \mathbb{R}^{H \times W}$$

1 channel

Decomposition-based fusion

$$\text{Fused} = \text{low-frequency (global)} + \text{high-frequency (edges)}$$

PCA-weighted

residual

METHOD – VII

For FB-PCA we calculate similar contribution scores as in METHOD - VI, but only for foreground(snowpole) pixels.

METHOD – VIII

Early gated Fusion

At the beginning of the Yolo pipeline, a convolution layer is added –

Apply convolution:

$$F = W_f * X$$

Dimensions

- $W_f \in \mathbb{R}^{64 \times 4 \times 3 \times 3}$
- Output:

$$F \in \mathbb{R}^{64 \times \frac{H}{2} \times \frac{W}{2}}$$

Another convolution produces a **gate map**:

$$G = \sigma(W_g * X)$$

Dimensions

- $W_g \in \mathbb{R}^{64 \times 4 \times 3 \times 3}$
- $G \in \mathbb{R}^{64 \times \frac{H}{2} \times \frac{W}{2}}$

Activation

$$\sigma = \text{sigmoid}$$

So:

$$G \in (0,1)$$

Gating

$$F_{\text{out}} = F \odot G$$

Dimensions

$$F_{\text{out}} \in \mathbb{R}^{64 \times \frac{H}{2} \times \frac{W}{2}}$$

Output channels – 64

METHOD – IX

Step 1: Feature extraction

Inside C3k2:

$$F = \mathcal{F}(X)$$

where $\mathcal{F}$ is a stack of conv + bottleneck layers.

Dimensions

$$F \in \mathbb{R}^{C \times H' \times W'}$$

Here:

- $C \approx 512$
- $H' = H/16, W' = W/16$

Step 2: Gate computation (late)

Now instead of using raw input, gate is computed from features:

$$G = \sigma(W_g * F)$$

Dimensions

- W_g : learnable conv
- $G \in \mathbb{R}^{C \times H' \times W'}$

Activation

$$\sigma = \text{Sigmoid} \Rightarrow G \in (0,1)$$

Step 3: Feature modulation

$$F_{\text{out}} = F \odot G$$